ALEXANDER MERCER

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Technical Skills

Languages: Python, TypeScript, Go, SQL, Bash, C++
Cloud & Infrastructure: AWS (ECS, Lambda, S3, RDS), Kubernetes, Docker, Terraform, CI/CD (GitHub Actions)
Frameworks & Libraries: FastAPI, Django, React, Next.js, Node.js, PyTorch, LangChain, Celery
Databases & Systems: PostgreSQL, Redis, Elasticsearch, Kafka, Apache Airflow, Snowflake
Practices & Tools: Distributed Systems, Microservices, RESTful APIs, GraphQL, TDD, Agile/Scrum

Professional Experience

Stripe — **San Francisco, CA**
Senior Software Engineer — Platform Infrastructure — *Aug 2023 – Present*

- Architected and delivered an asynchronous transaction ingestion pipeline handling 18M+ daily events, cutting peak processing latency by 42% while sustaining 99.995% service availability.
- Spearheaded Kubernetes cluster autoscaling optimization with Karpenter across 250+ microservices, reducing AWS compute expenditure by $320K annually.
- Mentored 5 mid-level engineers, established automated CI/CD security scanning gates, and authored end-to-end telemetry runbooks adopted across 4 platform teams.

DoorDash — **San Francisco, CA**
Software Engineer — Logistics & Dispatch — *Jun 2021 – Jul 2023*

- Engineered dynamic order-batching algorithms in Go and Redis, boosting courier dispatch efficiency by 14% across 12 high-density metropolitan markets.
- Migrated legacy monolithic dispatch workflows to distributed event-driven microservices using Apache Kafka, decreasing p99 API response times from 420ms to 85ms.
- Implemented comprehensive automated integration test suites with Pytest and Docker, increasing test coverage from 64% to 92% and preventing 3 critical deployment outages.

Vanguard FinTech — **New York, NY**
Junior Backend Engineer — *Jan 2020 – May 2021*

- Developed RESTful microservices in Python (FastAPI) and PostgreSQL to automate institutional portfolio rebalancing for 40,000+ active user accounts.
- Optimized complex SQL aggregation queries and implemented Redis caching, reducing database I/O bottlenecks by 38% during market opening hours.

Key Projects

NexusFlow: Distributed Task Orchestrator | *Go, gRPC, Raft, Docker, Prometheus* — **Jan 2024 – Mar 2024**

- Constructed a fault-tolerant distributed job scheduler leveraging Raft consensus to achieve zero-loss task failover across node outages.
- Benchmarked performance against Celery, achieving 3.5x higher throughput under synthetic 100K concurrent task burst workloads.

SemanticSearch: LLM Retrieval Engine | *Python, FastAPI, LangChain, Qdrant, React* — **Oct 2023 – Dec 2023**

- Engineered a retrieval-augmented generation (RAG) platform indexing 500K+ technical documents with sub-200ms hybrid vector search.
- Released as an open-source tool with 1,200+ GitHub stars; featured in top developer community newsletters.

Education

University of California, Berkeley — **Berkeley, CA**
Bachelor of Science in Computer Science; GPA: 3.84/4.00 — *Aug 2016 – Dec 2019*

- Honors: Dean's Honor List (6 Semesters), Upsilon Pi Epsilon Honor Society.
- Relevant Coursework: Distributed Systems, Algorithms & Data Structures, Operating Systems, Database Systems.

Certifications & Leadership

Certifications: AWS Certified Solutions Architect – Associate (2024), Certified Kubernetes Administrator (CKA, 2023)
Leadership: Open Source Contributor to Apache Arrow; Technical Speaker at SF Go Meetup (2024)